

Designing Cognitively-Aware Psychomotor Intelligent Tutoring Systems via Multi-Objective and Human-Centric Optimization

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Abstract—In this work, we present a methodology for integrating multiple tutoring objectives into model-based control policies for a psychomotor intelligent tutoring system (ITS). While tutoring objectives, like increasing self-efficacy, are well defined in the learning literature, realizing these types of objectives in an algorithmic framework is challenging. Motivated by the role of self-efficacy in psychomotor learning, we employ a Markov Decision Process (MDP) framework to train probabilistic models of self-confidence, workload, and learning stage. A custom reward function combines three key objectives: calibrating self-confidence with learning stage, calibrating workload to learning demands, and advancing learners from novice toward expert. Using this framework, we train two optimal tutoring policies—one prioritizing cognitive state calibration and the other focusing on learning progression. Using these two policies as test case scenarios, we conduct a between-subjects study and demonstrate how one can leverage the tunability of the multi-objective reward function to achieve desired learning outcomes.

I. INTRODUCTION

Social robots have been used in educational settings, largely to leverage their ability to interact with humans more naturally. They have been successful in positively impacting learning outcomes through shaping the learning process with personalized social interactions [1]. Their relevance and capabilities are further growing due to the integration of intelligent tutoring systems (ITS) in such robots [2], [1]. ITS personalize education for individual learners by emulating human tutors via algorithms that utilize human models and data collected from the human learner [3], [4]. In more conventional domains such as language learning, computer programming, or mathematics, ITS close the loop by utilizing human cognitive feedback via factors of self-confidence and workload [5]. According to seminal learning literature [6], human self-confidence and workload impact self-efficacy, or a learner’s self-awareness of their achievements. Furthermore, increasing student self-efficacy is correlated with better learning performance and higher student motivation [6]. Recently, there has been increasing interest in developing ITS for psychomotor tasks: any motor action based on conscious, mental activity [7], [8], [9]. However, ITS development for psychomotor tasks faces the following three challenges: (1) creating a knowledge space for the task, (2) maintaining

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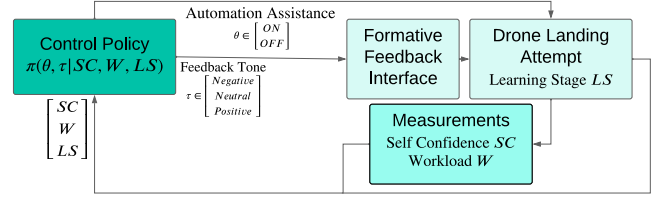


Fig. 1: The proposed model-based and multi-objective control framework.

learner motivation, and (3) personalizing the tutoring agent to learner characteristics [10]. Maintaining learner motivation requires online assessment of learner self-efficacy as well as controllable interventions that can influence it. Moreover, coordinating such an intervention with the overarching goal of improving learning in the context of psychomotor tasks is not straightforward due to difficulties in assessing ‘correct’ performance. A few psychomotor ITSs exist that try to address these challenges including SelfIt [10] and GIFT [8], predominantly through design emphasis on knowledge representation, as well as adaptive and self-improving tutors for better tutor-student interaction. There has been some focus on integrating learner “states”—namely engagement and motivation—into the design of these psychomotor ITS; however, these learner states can be further decomposed into the cognitive states of human self-confidence and workload. In turn, it is widely accepted that human cognitive states impact the quality of human-automation interactions [11], [12], [13]. Therefore, *psychomotor ITS design can be improved by leveraging an algorithmic framework that enables learning objectives (e.g., increased learner self-efficacy) to be codified through human cognitive states, learning progression representation, and adaptation to individual learners.* The contribution of this work is a **multi-input multi-output cognitive control framework for psychomotor ITS that can adaptively support the learner** by responding to cognitive states of self-confidence and workload as well as learning progression, visualized in Fig. 1. This framework builds on our prior work in which we developed an approach to classify user learning stage online [14] and demonstrated the benefits of calibrating learner self-confidence to their performance [15], [16]. However, our prior work was limited in considering only a single cognitive state and a single tutoring objective, despite the fact that for effective learning, multiple learning objectives must be balanced. Therefore, one focus of this work is to expand our learner model to include workload and learning stage, in addition to self-confidence, to address multiple learning objectives through reward design and control policy synthesis. We also extend the action space



Fig. 2: Web-deployable platform.

of the MDPs to include both automation assistance during task completion and tone of formative feedback after completion. Formative feedback is feedback provided to learners during the learning process to help learners self-assess their performance and understand actionable improvements [17]. In [18], Jensen et al. demonstrated how providing learners with AI-generated formative feedback using GPT4o [19] can improve learning outcomes; we take inspiration from their approach.

The paper is organized as follows. We describe the human user study in Section II. Next, training and validation of the learning stage, workload, and self-confidence modeling are discussed in Section III. In Section IV, we design, synthesize, and experimentally validate two optimal policies that each emphasize different objectives within our multi-objective reward function. We discuss the main conclusions and opportunities for future work in Section V.

II. HUMAN USER STUDY

In this work, we deploy a between-subjects user study in which participants learn how to safely land a quadrotor unassisted within 20 trials. The purpose of the study is two-fold—to first collect model training data that characterizes how users learn the task under various input conditions and then to test synthesized optimal control policies.

A. Experiment Design

In this experiment, the participant’s goal is to safely land the quadrotor onto a landing pad within appropriate constraints on quadrotor speed ($|v| \leq 15 \frac{m}{s}$) and roll angle ($|\phi| \leq 5^\circ$) in a desktop simulation shown in Fig. 2. The participants control the quadrotor using a keyboard. If the quadrotor reaches the landing pad but is outside of safe landing conditions, the participant achieves an unsafe landing. An unsuccessful landing occurs when the quadrotor crashes away from the landing pad. In each trial, the participant will either manually land the quadrotor or receive automation assistance. These two modes are called *manual* control mode (θ_{off}) and *shared* control mode (θ_{on}). In manual control mode, the participant fully controls the quadrotor inputs thrust force and roll angular acceleration. In shared control mode, the user’s input is augmented by the automation assistance: a linear quadratic regulator tracking a reference trajectory based on expert demonstrations using a linear time-invariant model of the quadrotor dynamics as well as a quadratic cost function that minimizes tracking error [20].

In between trials, participants receive formative feedback—feedback that is provided to a learner during a learning process—with components of *tone*, *improvement areas*, *actionable*, and *automation* [18]. The *tone* component influences the tone of the feedback given to the participant. The tone can be negative, neutral, or positive. The *improvement area* component makes sure that the generated feedback identifies improvement areas. The *actionable* component will suggest strategy changes to address the improvement area. The *automation* component ensures that the feedback reminds the participant that they received assistance in the last trial. Feedback is generated using GPT-4o [19]. For example:

“Your performance shows a good understanding of the drone’s operation, and you have made progress in navigating towards your landing target. The area where you can most improve is in controlling your landing speed. To improve in this area, increase thrust as needed to slow down when descending towards the landing pad. While you successfully managed the drone’s trajectory, it appears you received assistance during the landing, indicating that you may need more practice manually landing the drone.”

The experiment is deployed twice; what differs is the decision-making behind 1) when automation assistance is provided to the participant and 2) the tone of the formative feedback provided to the participant after each trial. For the first deployment of the experiment aimed at collecting open loop modeling data, the control mode for each trial, θ , is randomized between manual and shared mode, and the tone of feedback, τ , is randomized between negative, neutral, and positive; this results in a rich dataset well-suited for model training. In the second deployment of the experiment, we close the loop between the participant and the ITS so that θ and τ are determined by the trained optimal cognitive control policy $\pi(\theta, \tau)$. This occurs in trials 3-15; the first two and last five trials are completed in manual mode so that each participant’s manual performance can be quantified.

B. Participants

In total, 240 participants completed the user study on Prolific [21]. Of these participants, 22 were removed due to data collection errors (e.g. missing or too many trials) and poor quality data (such as never attempting to safely land or never adjusting both self-confidence and workload self-reports). The experiment was deployed twice; once for training and validation of models and once for implementing our trained policies. We will refer to these two datasets as “modeling data” and “policy data”, respectively.

The modeling data represents data collected from 178 participants. The data was divided such that approximately 80% of the data was used for training and the remainder used for testing. The model training dataset represents 143 participants (79 male, 64 female) with ages ranging 18-72 years ($\mu = 33.7$, $\sigma = 12.8$). The model testing dataset represents 35 participants (21 male, 14 female) with ages ranging 18-65

years ($\mu = 31.3$, $\sigma = 11.5$). In the second deployment, data from 40 participants (20 male, 20 female) were collected for closed-loop evaluation of the cognitive control policies (described in Section IV). Participant ages range from 19-75 years ($\mu = 36.5$ years, $\sigma = 12.7$). All participants were compensated \$8/hr. The Institutional Review Board at Purdue University, USA, approved the study.

C. Apparatus

The quadrotor landing training simulator was developed in Javascript and HTML. The simulator was originally developed for [20] and has been adapted and deployed for other human subjects experiments, published in [16] and [18]. The platform has been further adapted for the current work. A keyboard is used to control the quadrotor.

D. Procedure.

In total, each participant completes 20 trials of the quadrotor game. The learner can achieve one of four learning stages: (1) Novice, (2) Advanced Beginner, (3) Intermediate, and (4) Competent [14]. The learning stage is determined through a heuristic learning stage classifier adapted from [14]. After each trial k , participants are provided with the amount of time spent landing the quadrotor, landing speed, landing roll angle, whether they *unsuccessfully*, *unsafely*, or *safely* land the quadrotor, and what learning stage they have reached for all previous trials. They are subsequently asked to rate their self-confidence in their ability to land the quadrotor and their mental workload while performing the task, on a numerical scale of 0-100. The following definition for self-confidence is provided to each participant: “*The confidence in oneself and one’s powers and abilities.*” For workload, the participants are told that “*Workload is used to infer the mental demand of the task and mental demand refers to how much thinking, deciding, or calculating was required to perform the task.*”

III. HUMAN COGNITIVE STATE AND LEARNING PROGRESSION MODELING

To synthesize a cognitive control policy, we first train a data-driven probabilistic model of human cognitive states and learning progression (in terms of learning stage). Given the stochasticity of human behavior and prior work using Bayesian modeling techniques to model cognitive state dynamics, we assume the dynamics of interest behave like a Markov Decision Process (MDP). A MDP is defined as a 5-tuple $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma)$; the set $\mathcal{S}$ is a finite set of states and $\mathcal{A}$ is a finite set of actions. The transition probability function $\mathcal{T}$ is the probability of transitioning to the next state s' given the current state s and action a . The reward function $\mathcal{R}(s, a)$ represents the direct rewards associated with the agent’s decisions depending on the state and actions. Finally, the discount factor γ is a value between 0 and 1 and represents how important future rewards are at the current state. The reward function and discount factor are used to train a control policy given state dynamics. We use this MDP framework to capture the dynamics of cognitive states self-confidence and workload, as well as a performance state defined in

terms of learning stage. While it is likely that some dynamic coupling exists between the states, capturing this coupling in an MDP significantly increases the number of model parameters. Therefore, for this work, we assume each of the states is independent of the others such that independent models mapping the actions θ and τ to each individual state can be trained separately. Note that we support this assumption with validation results shown in Section III-B. It follows that the transition probability of all combinations of states can be calculated using self-confidence MDP, workload MDP, and learning stage MDP transition probabilities as given by Eqn. 1.

$$\mathcal{T}(s'|s, a) = \mathcal{T}(s'_{SC}|s_{SC}, a_\theta, a_\tau) \mathcal{T}(s'_W|s_W, a_\theta, a_\tau) \mathcal{T}(s'_{LS}|s_{LS}, a_\theta, a_\tau) \quad (1)$$

For the self-confidence MDP, the self-confidence state s_{SC} is discretized into low self-confidence SC_L , medium self-confidence SC_M , and high self-confidence SC_H . For workload MDP, the workload state s_W is discretized into low workload W_L , medium workload W_M , and high workload W_H . For both cognitive state models, the 33% and 66% quantiles of self-reported data on a numerical scale of 0-100 define thresholds for sorting each self-report into their respective discrete state. For the learning stage MDP, there are three learning stage states s_{LS} : learning stage 1 LS_1 , learning stage 2 LS_2 , and learning stages 3 and 4 $LS_{3,4}$. Learning stages 3 and 4 are combined as a design choice due to their similarities in quadrotor trajectory behavior. For all three MDP models, the set of actions consists of the decisions the tutoring agent can make. Specifically, the automation action a_θ represents if the within-trial automation assistance is off θ_{off} or on θ_{on} , and the feedback tone action a_τ represents if the feedback is negative τ_{neg} , neutral τ_{neu} , or positive τ_{pos} . Here, the states and actions were chosen to minimize the number of model parameters and maximize model predictive capability [16]; the definitions are summarized in Table I.

A. Model Training

The self-confidence MDP, workload MDP, and learning stage MDP model parameters are estimated by maximizing the likelihood of state sequences given the parameters. The model parameters are updated using the genetic algorithm in MATLAB’s Optimization Toolbox [22]. The state and action sequences are recorded in the model dataset. Therefore, the likelihood of state and action sequences is calculated using the joint probability of the series of states and actions up to trial k , i.e. $\mathbb{P}(s_{0:k}, a_{1:k})$. The chain rule of probability and Markov property [23], [24] are used to calculate the probability of state and action sequences such that

$$\mathbb{P}(s_{0:k}, a_{1:k}) = \mathbb{P}(s_0) \times \prod_{i=1}^k \mathbb{P}(s_i|a_i, s_{i-1}). \quad (2)$$

The likelihood of the state and action sequences until the last trial K is calculated using transition probabilities in Eqn. (1). The model is trained using a 5-fold cross-validation,

TABLE I: Definition of the human self-confidence, workload, and learning stage MDP models.

States $s \in \mathcal{S}$ = $\begin{bmatrix} \text{Self-Confidence } s_{SC} \\ \text{Workload } s_W \\ \text{Learning Stage } s_{LS} \end{bmatrix}$	$s_{SC} \in \mathcal{SC}$ $:= \begin{cases} \text{Low Self-Confidence } SC_L \\ \text{Medium Self-Confidence } SC_M \\ \text{High Self-Confidence } SC_H \end{cases}$
	$s_W \in \mathcal{W}$ $:= \begin{cases} \text{Low Workload } W_L \\ \text{Medium Workload } W_M \\ \text{High Workload } W_H \end{cases}$
	$s_{LS} \in \mathcal{LS}$ $:= \begin{cases} \text{Learning Stage 1 } LS_1 \\ \text{Learning Stage 2 } LS_2 \\ \text{Learning Stages 3 and 4 } LS_{3,4} \end{cases}$
Actions $a \in \mathcal{A}$ = $\begin{bmatrix} \text{Automation } a_\theta \\ \text{Tone } a_\tau \end{bmatrix}$	$a_\theta \in \theta$ $:= \begin{cases} \text{Assistance off } \theta_{off} \\ \text{Assistance on } \theta_{on} \end{cases}$
	$a_\tau \in \mathcal{T}$ $:= \begin{cases} \text{Negative } \tau_{neg} \\ \text{Neutral } \tau_{neu} \\ \text{Positive } \tau_{pos} \end{cases}$

wherein the modeling data is divided randomly into five equal sets or folds, and four folds are used for training. The model is trained five times using each fold once as test data. The final model is chosen based on which model achieves the highest log-likelihood for both the training and testing data set. A log-likelihood of -5.440×10^3 and -1.256×10^3 was achieved for the training and testing set, respectively. These values are summed from the individual log-likelihoods achieved by the self-confidence MDP (train: -1.489×10^3 , test: -339.7), workload MDP (train: -1.29×10^3 , test: -281.8), and learning stage MDP models (train: -2.661×10^3 , test: -634.8). The resulting transition probabilities for each model can be represented as a $|\mathcal{S}| \times |\mathcal{S}| \times |\mathcal{A}|$ matrix. That means that for the self-confidence, workload, and learning stage MDP models, a $3 \times 3 \times 6$ matrix captures all possible 54 state transitions. For demonstration, we provide an example of the transition probabilities $\mathcal{T}(SC'|SC, \theta_{on}, \tau_{pos})$ in Table II,

TABLE II: Transitions probability $\mathcal{T}(SC'|SC, a_\theta, a_\tau)$ where feedback tone τ is positive and assistance θ is both off (left) and on (right).

Actions States	θ_{off}, τ_{pos}			θ_{on}, τ_{pos}		
	SC'_L	SC'_M	SC'_H	SC'_L	SC'_M	SC'_H
SC_L	0.705	0.243	0.0528	0.781	0.153	0.0660
SC_M	0.207	0.632	0.161	0.176	0.725	0.0989
SC_H	0.121	0.110	0.769	0.115	0.244	0.642

B. Validation

The final model is validated to ensure that it is appropriate for optimal policy training. Multi-class receiver operating characteristic (ROC) curves are used to assess the predictive capabilities of each model in predicting self-confidence, workload, and learning stage of each participant. For each

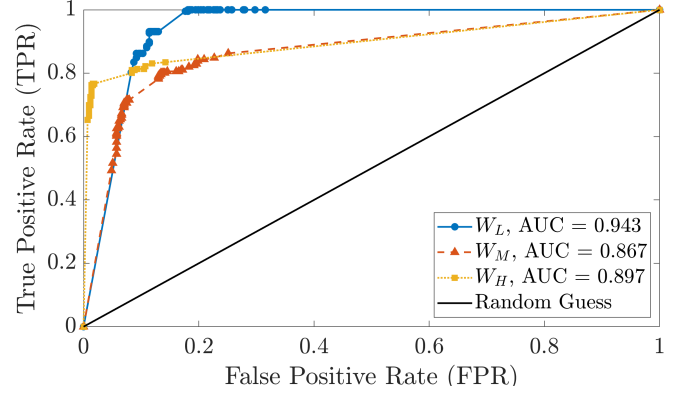


Fig. 3: One vs. rest ROC curves for workload.

ROC curve, the predicted cognitive states and learning stages are compared to the self-reported cognitive states and achieved learning stages, respectively. The true positive rate (TPR) and false positive rate (FPR) are plotted for threshold probabilities of 0-100% used to classify predicted self-confidence, workload, and learning stage.

The three discrete levels of self-confidence, workload, and learning stage serve as the different classes. Since ROC curves are generally used for binary classification, for each model, three ROC curves are generated using a one-vs-rest scheme and six ROC curves are generated using a one-vs-one scheme [25]. Fig. 3 shows an example of a one-vs-rest scheme for the workload MDP. In the one-vs-rest ROC, one class is treated as the positive class, and the rest are treated as negative classes. In the one-vs-one scheme, for each combination of two classes, data in any other class is removed and an ROC curve is generated using the remaining data. The areas under the curve (AUC) can be used to infer model predictive capability. A classifier based on random guess has an AUC of 0.5. A perfect classifier will have an AUC of 1. Typically, an AUC between 0.7 and 0.8 and between 0.8 and 0.9 is considered fair and good, respectively [26]. ROC curves are generated using 34 sets of participant test data, as mentioned in Section II-B. Table III lists all achieved AUCs for the one-vs-rest and one-vs-one ROC curves for all models. The lowest three AUCs are 0.503, 0.597, and 0.627, all occurring for ROC curves classifying LS_2 . This is likely due to LS_2 occurring the least out of all learning stage states in the modeling data. Otherwise, all AUCs are 0.694 and above.

IV. POLICY DESIGN, SYNTHESIS, AND EVALUATION

With trained MDP models describing the dynamic evolution of both cognitive states and learning progression as a function of controllable features of the ITS—automation assistance and formative feedback tone—we can design and synthesize an optimal policy to determine 1) when to assist the learner within trials and 2) what type of feedback to give the learner in between trials. We do this by formulating a multi-objective reward function based on tutoring objectives defined in terms of the modeled states.

TABLE III: Area under the curve (AUC) for all one-vs-one and one-vs-rest ROC curve classifications.

Self-Confidence Comparison	AUC	Workload Comparison	AUC	Learning Stage Comparison	AUC
SC_L vs SC_M	0.883	W_L vs W_M	0.916	LS_1 vs LS_2	0.729
SC_L vs SC_H	0.967	W_L vs W_H	0.968	LS_1 vs $LS_{3,4}$	0.830
SC_L vs rest	0.934	W_L vs rest	0.943	LS_1 vs rest	0.800
SC_M vs SC_L	0.884	W_M vs W_L	0.905	LS_2 vs LS_1	0.694
SC_M vs SC_H	0.771	W_M vs W_H	0.832	LS_2 vs $LS_{3,4}$	0.503
SC_M vs rest	0.805	W_M vs rest	0.867	LS_2 vs rest	0.597
SC_H vs SC_L	0.828	W_H vs W_L	0.906	$LS_{3,4}$ vs LS_1	0.797
SC_H vs SC_M	0.838	W_H vs W_M	0.888	$LS_{3,4}$ vs LS_2	0.627
SC_H vs rest	0.834	W_H vs rest	0.897	$LS_{3,4}$ vs rest	0.745

A. Reward Design

We aim to represent the following three tutoring objectives in the form of reward functions $R_i(s, a)$ for $i = 1, 2, 3$: (1) calibrate self-confidence to learning stage; (2) calibrate workload according to the Yerkes-Dodson law [27]; (3) reward forward progression in learning stage from novice toward expert. The first two tutoring objectives are grounded in the learning literature on self-efficacy [28], [29], [30], [31]. Increasing self-efficacy means that learners are more self-aware and realistic about their growth and achievements, leading to increased performance and motivation [31], [6]. Therefore, having appropriate self-confidence (with respect to performance) and appropriate workload are components of self-efficacy. The purpose of the third objective is to explicitly drive learners to progress (rather than regress) through the learning stages defined for this psychomotor task. The reward components designed based on the three tutoring objectives will be referred to as objectives 1, 2, and 3, respectively.

Objective 1: Reward function $R_1(s, a)$ is designed with equal weighting between self-confidence calibration using the automation assistance action as feedback $R(s_{SC}, s_{LS}, a_\theta)$ and self-confidence calibration based on the feedback tone action $R(s_{SC}, s_{LS}, a_\tau)$, as depicted in Table IV. The reward is designed to drive the discrete states of self-confidence to match that of the learning stages. For example, for $R(s_{SC}, s_{LS}, a_\tau)$ (see Table IV), when the learner displays severe overconfidence by reporting high self-confidence despite only achieving learning stage 1, a reward of 0.60 is associated with negative toned feedback, which is higher than the reward of 0.20 associated with positive toned feedback. The reward associated with automation not assisting is higher than its counterpart of assisting so that the learner can be shown that they perform worse without assistance. The final rewards for R_1 are shown in Table V.

Objective 2: Reward function $R_2(s, a)$ is designed to drive mental workload to appropriate levels according to the achieved learning stage. This is supported by learning stage theory from [32] in which the mental demand from completing a task decreases as humans gain expertise. This has also been observed in our prior work when characterizing learning stage definitions and developing the learning stage classifier for our psychomotor task [14]. This is reflected in the reward design shown in Table VI. When the learner achieves learning stage 3 or 4, a reward of 1.0 is associated

TABLE IV: Rewards function components $R(s_{SC}, s_{LS}, a_\theta)$ and $R(s_{SC}, s_{LS}, a_\tau)$ for calibrating self-confidence to learning stage. Green highlights represent cases of calibration, and yellow and red highlights represent cases of light and severe mis-calibration, respectively.

s_{LS}	s_{SC}	$R(s_{SC}, s_{LS}, a_\theta)$		$R(s_{SC}, s_{LS}, a_\tau)$		
		θ_{off}	θ_{on}	τ_{neg}	τ_{neu}	τ_{pos}
LS_1	SC_L	0.75	1.0	0.10	0.50	0.40
	SC_M	0.75	0.50	0.50	0.30	0.20
	SC_H	0.50	0.25	0.60	0.30	0.20
LS_2	SC_L	0.50	0.75	0.20	0.30	0.50
	SC_M	0.75	0.75	0.10	0.50	0.40
	SC_H	0.75	0.50	0.50	0.30	0.20
$LS_{3,4}$	SC_L	0.25	0.50	0.10	0.30	0.60
	SC_M	0.50	0.75	0.10	0.40	0.50
	SC_H	1.0	0.75	0.10	0.50	0.40

TABLE V: Reward $R_1(s, a)$ for calibrating self-confidence to learning stage. Green highlights represent cases of calibration, and yellow and red highlights represent cases of light and severe mis-calibration, respectively.

	TONE a_τ :	τ_{neg}		τ_{neu}		τ_{pos}	
		θ_{off}	θ_{on}	θ_{off}	θ_{on}	θ_{off}	θ_{on}
LS_1	SC_L	0.43	0.65	0.78	1.00	0.70	0.91
	SC_M	0.78	0.57	0.61	0.39	0.52	0.30
	SC_H	0.65	0.43	0.39	0.17	0.22	0.010
LS_2	SC_L	0.30	0.52	0.39	0.61	0.57	0.78
	SC_M	0.44	0.43	0.79	0.77	0.70	0.69
	SC_H	0.78	0.57	0.61	0.39	0.52	0.30
$LS_{3,4}$	SC_L	0.010	0.22	0.17	0.39	0.43	0.65
	SC_M	0.22	0.43	0.48	0.70	0.57	0.78
	SC_H	0.65	0.43	1.00	0.78	0.91	0.70

with low workload and no assistance. Again, this is because a main quality of an expert includes the ability to safely land the drone manually with low mental demand. On the other hand, if the learner is in learning stage 1, a reward of 1.0 is associated with low workload because the learner is a novice, and therefore attempting to land the quadrotor manually will be difficult and increase workload to more appropriate levels. If the learner's current workload is high, a higher reward is given for the automation assistance in comparison to not assisting, as the user needs help and does not have the mental demand capacity to complete the task without assistance. We choose to have the rewards depend

only on the assistance action and not the feedback tone action since assistance is given *within trials*, whereas the feedback is given after attempting the task.

TABLE VI: Reward $R_2(s, a)$ for calibrating workload. Green, yellow, and red highlights represent cases of appropriate workload, light under- and over-workload, and severe under- and over-workload, respectively.

	ASSIST a_θ :	θ_{off}	θ_{on}
LS_1	W_L	1.0	0.010
	W_M	0.60	0.40
	W_H	0.30	0.70
LS_2	W_L	0.60	0.40
	W_M	0.45	0.55
	W_H	0.40	0.60
$LS_{3,4}$	W_L	1.0	0.010
	W_M	0.70	0.30
	W_H	0.40	0.60

Objective 3: Reward function $R_3(s, a)$ is designed to reward participants when they achieve advanced stages of learning. Table VII shows the rewards. The rewards are designed to encourage the learner to reach higher learning stages in manual mode and provide assistance when achieving lower learning stages. It follows that a high reward of 1.0 is given when the user achieves learning stage 3 or 4 with no assistance, and the reward is decreased for learning stages 1 or 2. If the automation assistance is on, a high reward of 1.0 is assigned for the learning stage 1 state as it is likely the user needs help within the trial. The reward decreases for learning stages 2-4 as the user is less likely to need assistance. We again choose to have the rewards depend only on the assistance action and not the feedback tone action since the assistance action directly impacts the user's performance within each trial.

TABLE VII: Reward $R_3(s, a)$ for advancing learning stage.

ASSIST a_θ :	θ_{off}	θ_{on}
LS_1	0.10	1.0
LS_2	0.55	0.45
$LS_{3,4}$	1.0	0.25

The reward function components are designed such that all rewards range between 0 and 1. Two comprehensive reward functions are generated by convex combination of the reward functions R_1 , R_2 and R_3 :

$$\bar{\mathcal{R}}(s, a) = \alpha_1 R_1 + \alpha_2 R_2 + \alpha_3 R_3, \quad (3)$$

in which α_1 , α_2 and α_3 are the weights associated with the prioritization of each tutoring objective. For demonstration purposes, we consider two different weightings of rewards. The first comprehensive reward $\bar{\mathcal{R}}_1(s, a)$ will weigh tutoring objectives 1 and 2 higher than objective 3 such that $\bar{\mathcal{R}}_1(s, a) = 0.4R_1 + 0.4R_2 + 0.2R_3$. This reward prioritizes **driving learners' cognitive states to appropriate levels**

throughout the learning process. The second reward will be the opposite in which $\bar{\mathcal{R}}_2(s, a) = 0.1R_1 + 0.1R_2 + 0.8R_3$ so that the tutoring objective of **advancing learning stage is prioritized the most.**

B. Policy Training

Optimal policies are trained using the Bellman optimality equations [23], [24]. The action-value function $Q(s, a)$ is calculated using

$$Q(s, a) = \mathcal{R}(s, a) + \gamma \sum_{s' \in \mathcal{S}} \mathcal{T}(s'|s, a) \sum_{a' \in \mathcal{A}} Q(s', a') \quad (4)$$

where $\mathcal{T}(s'|s, a)$ and $\mathcal{R}(s, a)$ are from Eqn. 1 and Eqn. 3, respectively. From reward functions $\bar{\mathcal{R}}_1(s, a)$ and $\bar{\mathcal{R}}_2(s, a)$, action-value functions $Q_1(s, a)$ and $Q_2(s, a)$ are trained using value iteration, respectively. The optimal policies $\pi_1(a|s)$ and $\pi_2(a|s)$ are then determined by the action associated with the largest value for each state combination in $Q_1(s, a)$ and $Q_2(s, a)$, respectively. The policies are shown in Fig. 4 and, as expected, the policies are different due to each reward function design. For example, the optimal policy $\pi_1(a|s)$ can provide assistance in all learning stage states whereas $\pi_2(a|s)$, which strongly prioritizes progression in learning stage, only provides assistance in learning stage 1.

C. Implementation

We redeployed the experiment described in Section II using the two policies $\pi_1(a|s)$ and $\pi_2(a|s)$, which we will refer to as treatment 1 and treatment 2, respectively. We collected 20 sets of participant data for each treatment. Independent t -tests were applied to the number of learning stage 3 and 4 achievements per participant in the last five trials, self-reported self-confidence in the last five trials, and self-reported workload in all trials as a comparison between the two treatments. Recall that the first two and last five trials are completed without automation assistance. Box plots for each comparison are provided in Fig. 5.

Advanced Learning Stage: From Fig. 5, we see that those who received treatment 2 achieved learning stages 3 and 4 in the last five trials more than those who received treatment 1; this is expected given the higher weighting on R_3 in treatment 2. However, the difference in number of achieved learning stages 3 and 4 between treatment 1 ($\mu = 8.40$, $\sigma = 4.27$) and treatment 2 ($\mu = 10.7$, $\sigma = 3.56$) was not found to be significant ($t(38) = -1.85$, $p = 0.0720$).

Self-confidence: The distribution of self-confidence data shown in Fig. 5 shows that learners who received treatment 2 had higher self-confidence ($\mu = 44.7$, $\sigma = 24.1$) compared to learners who received treatment 1 ($\mu = 60.1$, $\sigma = 28.0$) in the last five trials. This difference is significant ($t(198) = -4.18$, $p = 4.38 \times 10^{-5}$). Learners who received treatment 2 achieved learning stages 3 and 4 more than those who received treatment 1, which may contribute to learners receiving treatment 2 reporting higher self-confidence. Moreover, $\pi_1(a|s)$ in Fig. 4 was trained to weigh calibrating self-confidence more, so learners receiving treatment 1 (who achieved learning stages 1 and 2 more than those receiving



Fig. 4: The resulting optimal cognitive control policies $\pi_1(a|s)$ and $\pi_2(a|s)$ from action value function $Q_1(s, a)$ and $Q_2(s, a)$, respectively. Recall that self-reported self-confidence and workload are reported on a scale of 0-100 and the 33% and 66% quantiles serve as thresholds (represented by black lines) for low, medium, and high discretization.

treatment 2) likely experienced more negative feedback and more cases of no assistance from the automation.

Workload: Learners who received treatment 2 report significantly lower workload ($\mu = 64.1$, $\sigma = 25.7$) compared to learners who received treatment 1 ($\mu = 71.0$, $\sigma = 26.4$) ($t(798) = 3.72$, $p = 2.14 \times 10^{-8}$). It can be seen in Fig. 5 that the 25% and 75% quantiles are much tighter for treatment 2 than the corresponding quantiles in treatment 1. This indicates that learners receiving treatment 1 report a wider range of workload in comparison to those receiving treatment 2. This is likely due to training $\pi_1(a|s)$ to more heavily weigh driving workload to appropriate levels from objective 2 in Section IV-A in comparison to that of $\pi_2(a|s)$. Workload is more actively and purposely influenced in treatment 1 in comparison to treatment 2.

In summary, learners from both treatments achieve similar learning stage outcomes at the end of the 20 trials, but learners receiving treatment 1 experience lower self-confidence and a larger range of workload than those receiving treatment 2. From this, the two policies trained using different prioritization of the tutoring objectives resulted in different learning outcomes. Individual learners may benefit from personalization via different weightings of the tutoring objectives. This would result in personalized policies and will be investigated in future work.

V. CONCLUSION

In this work we presented a control-centric methodology for embedding multiple tutoring objectives into a psychomotor intelligent tutoring system. This multi-objective, human-centered approach is driven by the importance of self-efficacy toward achieving learning outcomes, particularly for psychomotor tasks. Importantly, the proposed approach uses a generalizable model framework and parameter definitions such that it is broadly applicable.

A central component of our approach is a custom reward function that balances three key tutoring objectives: (1) aligning self-confidence with learning stage, (2) calibrating workload to learning stage, and (3) promoting forward progression through learning stages. Using three independent MDP models and the proposed multi-objective reward function, we trained optimal control policies for an intelligent tutoring agent capable of adapting to changes in a learner’s self-confidence, workload, and learning stage. To demonstrate the tunability of the multi-objective reward function, we designed and evaluated two different policies: one prioritizing cognitive state calibration to participants’ learning stage and another emphasizing progression through learning stages. The two policies using different prioritization of the tutoring objectives resulted in significantly different learning outcomes when implemented.

Future work will explore inter-dependencies between cognitive states and learning stages, as well as investigate alternative modeling approaches to better capture the input-output dynamics of tutoring interactions. Additionally, we plan to enhance the personalization of policies by experimenting with alternative weightings and prioritization of tutoring objectives within the reward function for individual learners. Ultimately, we aim to improve the embedded intelligence of social robots used for education through increased quality of instruction and personalization to human learners, thereby improving human-automation interaction experience for human learners.

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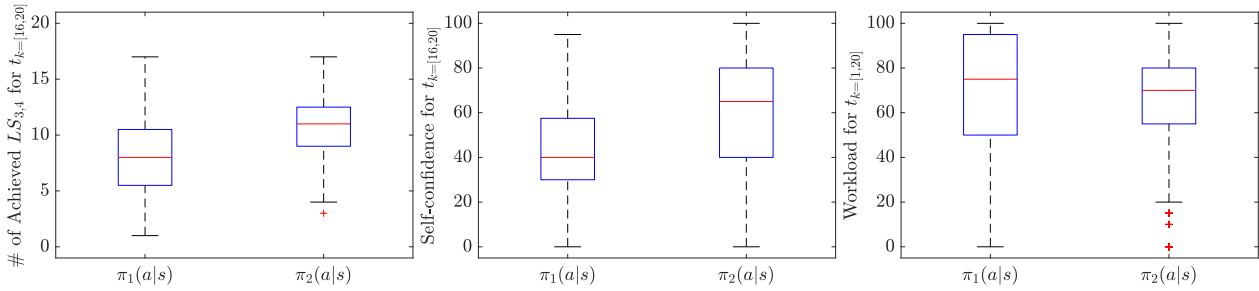


Fig. 5: Box plots comparing number of achieved learning stages 3 and 4 in trials 16-20, self-report self-confidence in trials 16-20, and self-report workload in all trials between policies $\pi_1(a|s)$ and $\pi_2(a|s)$.

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